

Monte Carlo Replication of DML-IVQR

Comparison with Chen, Huang and Tien

Computational environment

Authors' environment

Component	Version
R	3.4.3
quantreg	5.34
hdm	0.2.0
hqreg	1.4
mvtnorm	1.0-6
doSNOW	1.0.16

Replication environment

Component	Version
R	4.6.1
quantreg	6.1
hdm	0.3.2
hqreg	1.4.1
mvtnorm	1.4.2
doSNOW	1.0.20

Monte Carlo design

The Monte Carlo experiment evaluates whether DML-IVQR can recover quantile treatment effects in a high-dimensional setting with many control variables.

The simulation uses two sample sizes:

- $n = 500$
- $n = 1000$

For each sample size, 500 independent Monte Carlo replications are conducted. This gives 1,000 simulated samples in total.

Each simulated dataset contains:

- one outcome variable, Y_i ;
- one endogenous treatment variable, D_i ;
- two instruments, Z_{1i} and Z_{2i} ;
- $p = 100$ control variables, $X_{1i}, \dots, X_{100i}$.

Only a small subset of the controls is relevant. The first seven controls enter the outcome equation directly, while ten controls are included in the oracle specification because some of them also enter the instrument equations. The remaining controls are irrelevant noise variables. The design therefore creates a high-dimensional but sparse model.

Data-generating process

The two structural disturbances are jointly normally distributed:

$$\begin{pmatrix} u_i \\ \varepsilon_i \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 0.3 \\ 0.3 & 1 \end{pmatrix} \right).$$

The correlation of 0.3 between u_i and ε_i creates endogeneity.

The latent control variables and instrument shocks are independently drawn from standard normal distributions:

$$x_{ji}, z_{1i}, z_{2i}, v_{1i}, v_{2i} \sim N(0, 1).$$

The observed control variables are transformed to the unit interval:

$$X_{ji} = \Phi(x_{ji}),$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function.

The treatment variable is generated as

$$D_i = \Phi(z_{1i} + z_{2i} + \varepsilon_i).$$

The two instruments are constructed as

$$Z_{1i} = z_{1i} + X_{2i} + X_{3i} + X_{4i} + v_{1i},$$

and

$$Z_{2i} = z_{2i} + X_{7i} + X_{8i} + X_{9i} + X_{10i} + v_{2i}.$$

The outcome equation is

$$Y_i = 1 + D_i + 5X_{1i} + 5X_{2i} + 5X_{3i} + 5X_{4i} + 5X_{5i} + 5X_{6i} + 5X_{7i} + D_i u_i.$$

Because D_i is multiplied by u_i , the treatment effect varies across the outcome distribution.

True quantile treatment effect

The true structural effect at quantile τ is

$$\alpha_0(\tau) = 1 + \Phi^{-1}(\tau).$$

The experiment evaluates

$$\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}.$$

The corresponding true values are approximately:

- $\alpha_0(0.10) = -0.282$
- $\alpha_0(0.25) = 0.326$
- $\alpha_0(0.50) = 1.000$
- $\alpha_0(0.75) = 1.674$
- $\alpha_0(0.90) = 2.282$

For each quantile, the estimator searches over the grid

$$\alpha \in [-1, 3]$$

in increments of 0.1.

Estimators

The simulation compares four estimators:

- **res-GMM:** oracle GMM using the ten relevant controls and residualizing the instruments on the controls;
- **GMM:** oracle GMM using the ten relevant controls without residualizing the instruments;
- **full-GMM:** GMM using all 100 controls without regularization;
- **DML-IVQR:** the proposed regularized estimator for high-dimensional IVQR.

Performance measures

Estimator performance is evaluated using root-mean-square error, mean absolute error, and bias.

The root-mean-square error is

$$\text{RMSE} = \sqrt{\frac{1}{R} \sum_{r=1}^R (\hat{\alpha}_r - \alpha_0)^2}.$$

The mean absolute error is

$$\text{MAE} = \frac{1}{R} \sum_{r=1}^R |\hat{\alpha}_r - \alpha_0|.$$

Bias follows the sign convention used in the paper:

$$\text{BIAS} = \alpha_0 - \bar{\hat{\alpha}}.$$

Purpose of the comparisons

The three simulation tables answer different questions:

- **Table 1:** Does residualizing the instruments on the controls improve efficiency?
- **Table 2:** Does DML-IVQR outperform unregularized full-GMM and approach the oracle-GMM benchmark?
- **Table 3:** How does the theoretically motivated Belloni-Chernozhukov penalty compare with five-fold cross-validation?

Table 1: Residualizing and non-residualizing Z on X

Authors

Model	n=500 RMSE	n=500 MAE	n=500 BIAS	n=1000 RMSE	n=1000 MAE	n=1000 BIAS
alpha0.10 (res-GMM)	0.1888	0.1510	-0.0894	0.1219	0.0950	-0.0552
alpha0.10 (GMM)	0.4963	0.2559	-0.1776	0.1632	0.1138	-0.0628
alpha0.25 (res-GMM)	0.1211	0.0967	-0.0335	0.0812	0.0654	-0.0257
alpha0.25 (GMM)	0.1783	0.1179	-0.0255	0.0964	0.0754	-0.0235
alpha0.50 (res-GMM)	0.0990	0.0716	0.0092	0.0690	0.0436	-0.0020
alpha0.50 (GMM)	0.1437	0.1016	0.0340	0.0801	0.0542	0.0078
alpha0.75 (res-GMM)	0.1374	0.1066	0.0553	0.0828	0.0676	0.0213
alpha0.75 (GMM)	0.2403	0.1710	0.1295	0.1147	0.0849	0.0443
alpha0.90 (res-GMM)	0.2438	0.1839	0.1226	0.1391	0.1067	0.0668
alpha0.90 (GMM)	0.8483	0.5341	0.4960	0.3481	0.1967	0.1614

Replication

Model	n=500 RMSE	n=500 MAE	n=500 BIAS	n=1000 RMSE	n=1000 MAE	n=1000 BIAS
alpha0.10 (res-GMM)	0.2001	0.1580	-0.1096	0.1193	0.0934	-0.0558
alpha0.10 (GMM)	0.4143	0.2473	-0.1770	0.1679	0.1186	-0.0732
alpha0.25 (res-GMM)	0.1163	0.0939	-0.0389	0.0782	0.0642	-0.0225
alpha0.25 (GMM)	0.1693	0.1157	-0.0395	0.0886	0.0700	-0.0169
alpha0.50 (res-GMM)	0.1028	0.0728	0.0000	0.0660	0.0404	-0.0020
alpha0.50 (GMM)	0.1472	0.1026	0.0206	0.0846	0.0544	0.0120
alpha0.75 (res-GMM)	0.1335	0.1047	0.0349	0.0824	0.0673	0.0175
alpha0.75 (GMM)	0.2344	0.1599	0.1001	0.1113	0.0860	0.0417
alpha0.90 (res-GMM)	0.2731	0.1966	0.1322	0.1313	0.1016	0.0626
alpha0.90 (GMM)	0.8790	0.5513	0.5142	0.3017	0.1901	0.1602

Comparison. Both sets of results show that residualizing the instruments reduces RMSE and MAE relative to non-residualized GMM, with the largest gains at the tail quantiles and in the smaller sample.

Table 2: IVQR with high-dimensional controls

Authors

Sample	Model	RMSE (ratio)	MAE (ratio)	BIAS
n = 500	alpha0.10 (full-GMM)	0.7648 (4.05)	0.6646 (4.40)	-0.6534
n = 500	alpha0.10 (oracle-GMM)	0.1888 (1.00)	0.1510 (1.00)	-0.0894
n = 500	alpha0.10 (DML-IVQR)	0.3113 (1.65)	0.2389 (1.58)	-0.2040
n = 500	alpha0.25 (full-GMM)	0.2712 (2.24)	0.2213 (2.29)	-0.1877
n = 500	alpha0.25 (oracle-GMM)	0.1211 (1.00)	0.0967 (1.00)	-0.0335
n = 500	alpha0.25 (DML-IVQR)	0.1562 (1.29)	0.1254 (1.30)	-0.0797
n = 500	alpha0.50 (full-GMM)	0.1628 (1.64)	0.1234 (1.72)	0.0190
n = 500	alpha0.50 (oracle-GMM)	0.0990 (1.00)	0.0716 (1.00)	0.0092
n = 500	alpha0.50 (DML-IVQR)	0.1169 (1.18)	0.0846 (1.18)	-0.0186
n = 500	alpha0.75 (full-GMM)	0.3422 (2.49)	0.2806 (2.63)	0.2503
n = 500	alpha0.75 (oracle-GMM)	0.1374 (1.00)	0.1066 (1.00)	0.0553
n = 500	alpha0.75 (DML-IVQR)	0.1495 (1.09)	0.1167 (1.09)	0.0517
n = 500	alpha0.90 (full-GMM)	0.9449 (3.88)	0.8032 (4.37)	0.7892
n = 500	alpha0.90 (oracle-GMM)	0.2438 (1.00)	0.1839 (1.00)	0.1226
n = 500	alpha0.90 (DML-IVQR)	0.3568 (1.46)	0.2609 (1.42)	0.2012
n = 1000	alpha0.10 (full-GMM)	0.3918 (3.21)	0.3442 (3.62)	-0.3304
n = 1000	alpha0.10 (oracle-GMM)	0.1219 (1.00)	0.0950 (1.00)	-0.0552
n = 1000	alpha0.10 (DML-IVQR)	0.1376 (1.13)	0.1085 (1.14)	-0.0760
n = 1000	alpha0.25 (full-GMM)	0.1646 (2.03)	0.1361 (2.08)	-0.1135
n = 1000	alpha0.25 (oracle-GMM)	0.0812 (1.00)	0.0654 (1.00)	-0.0257
n = 1000	alpha0.25 (DML-IVQR)	0.0992 (1.22)	0.0804 (1.23)	-0.0437
n = 1000	alpha0.50 (full-GMM)	0.1038 (1.50)	0.0754 (1.73)	-0.0002
n = 1000	alpha0.50 (oracle-GMM)	0.0690 (1.00)	0.0436 (1.00)	-0.0020
n = 1000	alpha0.50 (DML-IVQR)	0.0776 (1.12)	0.0510 (1.17)	-0.0142
n = 1000	alpha0.75 (full-GMM)	0.1747 (2.11)	0.1452 (2.15)	0.1175
n = 1000	alpha0.75 (oracle-GMM)	0.0828 (1.00)	0.0676 (1.00)	0.0213
n = 1000	alpha0.75 (DML-IVQR)	0.0930 (1.12)	0.0741 (1.10)	0.0227
n = 1000	alpha0.90 (full-GMM)	0.4320 (3.11)	0.3682 (3.45)	0.3496
n = 1000	alpha0.90 (oracle-GMM)	0.1391 (1.00)	0.1067 (1.00)	0.0668
n = 1000	alpha0.90 (DML-IVQR)	0.1649 (1.19)	0.1232 (1.15)	0.0732

Replication

Sample	Model	RMSE (ratio)	MAE (ratio)	BIAS
n = 500	alpha0.10 (full-GMM)	0.7643 (3.82)	0.6680 (4.23)	-0.6594
n = 500	alpha0.10 (oracle-GMM)	0.2001 (1.00)	0.1580 (1.00)	-0.1096
n = 500	alpha0.10 (DML-IVQR)	0.3086 (1.54)	0.2426 (1.54)	-0.2126
n = 500	alpha0.25 (full-GMM)	0.2927 (2.52)	0.2458 (2.62)	-0.2091
n = 500	alpha0.25 (oracle-GMM)	0.1163 (1.00)	0.0939 (1.00)	-0.0389
n = 500	alpha0.25 (DML-IVQR)	0.1613 (1.39)	0.1294 (1.38)	-0.0895
n = 500	alpha0.50 (full-GMM)	0.1737 (1.69)	0.1318 (1.81)	0.0054
n = 500	alpha0.50 (oracle-GMM)	0.1028 (1.00)	0.0728 (1.00)	0.0000
n = 500	alpha0.50 (DML-IVQR)	0.1196 (1.16)	0.0870 (1.20)	-0.0202
n = 500	alpha0.75 (full-GMM)	0.3312 (2.48)	0.2668 (2.55)	0.2297
n = 500	alpha0.75 (oracle-GMM)	0.1335 (1.00)	0.1047 (1.00)	0.0349
n = 500	alpha0.75 (DML-IVQR)	0.1484 (1.11)	0.1160 (1.11)	0.0333
n = 500	alpha0.90 (full-GMM)	0.9014 (3.30)	0.7646 (3.89)	0.7422
n = 500	alpha0.90 (oracle-GMM)	0.2731 (1.00)	0.1966 (1.00)	0.1322
n = 500	alpha0.90 (DML-IVQR)	0.3747 (1.37)	0.2621 (1.33)	0.2138
n = 1000	alpha0.10 (full-GMM)	0.3783 (3.17)	0.3311 (3.54)	-0.3206
n = 1000	alpha0.10 (oracle-GMM)	0.1193 (1.00)	0.0934 (1.00)	-0.0558
n = 1000	alpha0.10 (DML-IVQR)	0.1379 (1.16)	0.1093 (1.17)	-0.0754
n = 1000	alpha0.25 (full-GMM)	0.1654 (2.11)	0.1336 (2.08)	-0.1083
n = 1000	alpha0.25 (oracle-GMM)	0.0782 (1.00)	0.0642 (1.00)	-0.0225
n = 1000	alpha0.25 (DML-IVQR)	0.0920 (1.18)	0.0741 (1.15)	-0.0419
n = 1000	alpha0.50 (full-GMM)	0.1037 (1.57)	0.0748 (1.85)	0.0004
n = 1000	alpha0.50 (oracle-GMM)	0.0660 (1.00)	0.0404 (1.00)	-0.0020
n = 1000	alpha0.50 (DML-IVQR)	0.0763 (1.16)	0.0498 (1.23)	-0.0146
n = 1000	alpha0.75 (full-GMM)	0.1729 (2.10)	0.1438 (2.14)	0.1219
n = 1000	alpha0.75 (oracle-GMM)	0.0824 (1.00)	0.0673 (1.00)	0.0175
n = 1000	alpha0.75 (DML-IVQR)	0.0898 (1.09)	0.0721 (1.07)	0.0213
n = 1000	alpha0.90 (full-GMM)	0.4386 (3.34)	0.3733 (3.67)	0.3626
n = 1000	alpha0.90 (oracle-GMM)	0.1313 (1.00)	0.1016 (1.00)	0.0626
n = 1000	alpha0.90 (DML-IVQR)	0.1614 (1.23)	0.1204 (1.19)	0.0710

Comparison. In both the authors' results and the replication, DML-IVQR substantially outperforms unregularized full-GMM and lies much closer to the infeasible oracle-GMM benchmark. RMSE generally decreases when the sample size rises from 500 to 1000.

Table 3: Choice of penalty

Authors

Model	n=500 RMSE	n=500 MAE	n=500 BIAS	n=1000 RMSE	n=1000 MAE	n=1000 BIAS
alpha0.25 (Belloni and Chernozhukov)	0.1716	0.1325	-0.0716	0.0849	0.0683	0.0056
alpha0.25 (5-fold Cross-Validation)	0.1720	0.1368	-0.0986	0.0995	0.0811	-0.0589
alpha0.50 (Belloni and Chernozhukov)	0.1273	0.0962	0.0270	0.0800	0.0556	0.0384
alpha0.50 (5-fold Cross-Validation)	0.1374	0.1032	-0.0384	0.0779	0.0536	-0.0236
alpha0.75 (Belloni and Chernozhukov)	0.1572	0.1272	0.0876	0.1142	0.0961	0.0839
alpha0.75 (5-fold Cross-Validation)	0.1526	0.1179	0.0286	0.0838	0.0677	0.0205

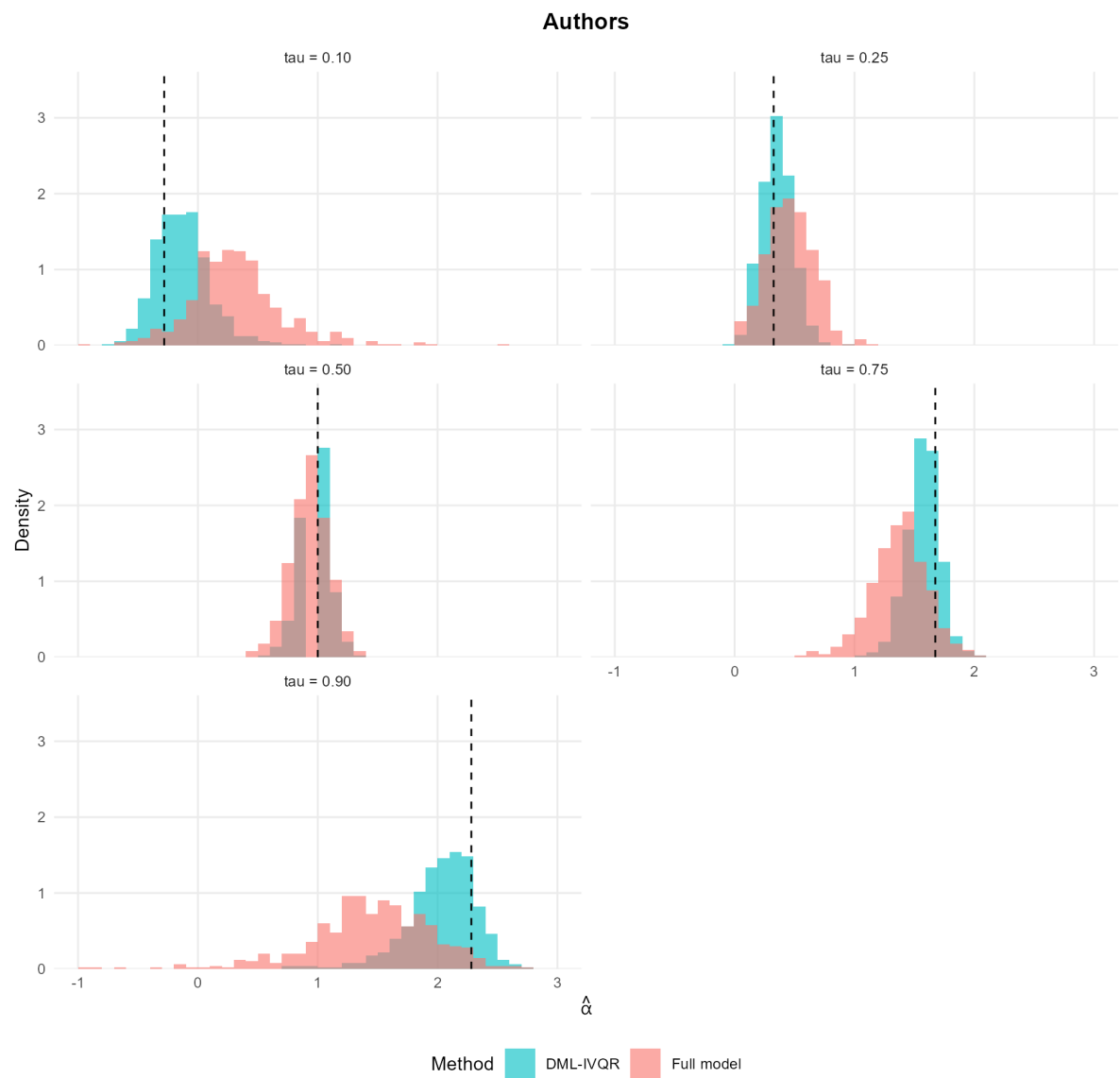
Replication

Model	n=500 RMSE	n=500 MAE	n=500 BIAS	n=1000 RMSE	n=1000 MAE	n=1000 BIAS
alpha0.25 (Belloni and Chernozhukov)	0.1490	0.1182	0.0151	0.0902	0.0725	0.0385
alpha0.25 (5-fold Cross-Validation)	0.1651	0.1299	-0.0851	0.0961	0.0792	-0.0541
alpha0.50 (Belloni and Chernozhukov)	0.1393	0.1100	0.0688	0.0921	0.0668	0.0560
alpha0.50 (5-fold Cross-Validation)	0.1182	0.0876	-0.0224	0.0727	0.0476	-0.0164
alpha0.75 (Belloni and Chernozhukov)	0.1847	0.1502	0.1345	0.1264	0.1081	0.0995
alpha0.75 (5-fold Cross-Validation)	0.1515	0.1180	0.0511	0.0871	0.0703	0.0271

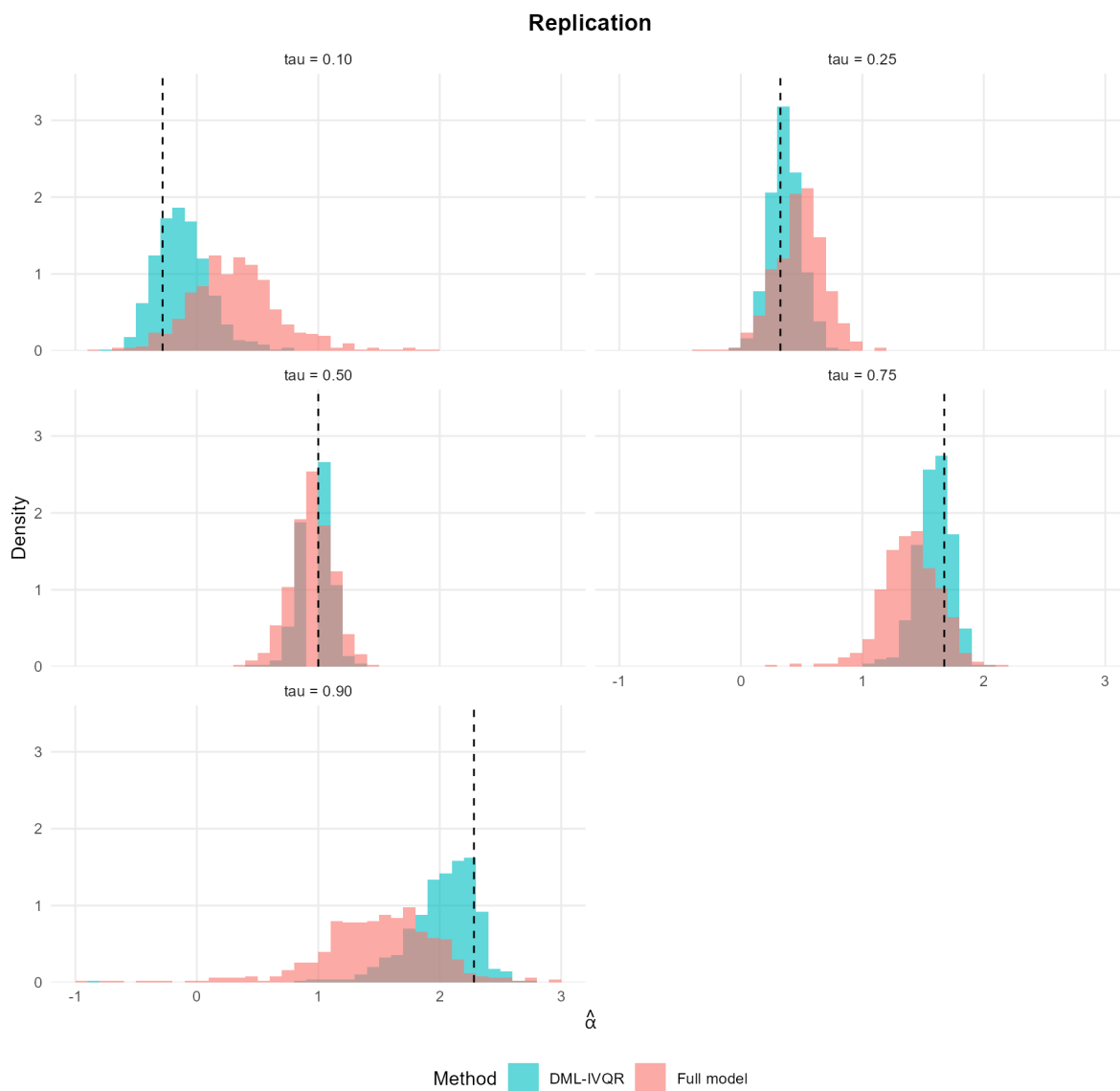
Comparison. The Belloni-Chernozhukov and five-fold cross-validation procedures produce broadly similar finite-sample performance in both sets of results.

Figure 1: Distribution of estimates

Authors



Replication

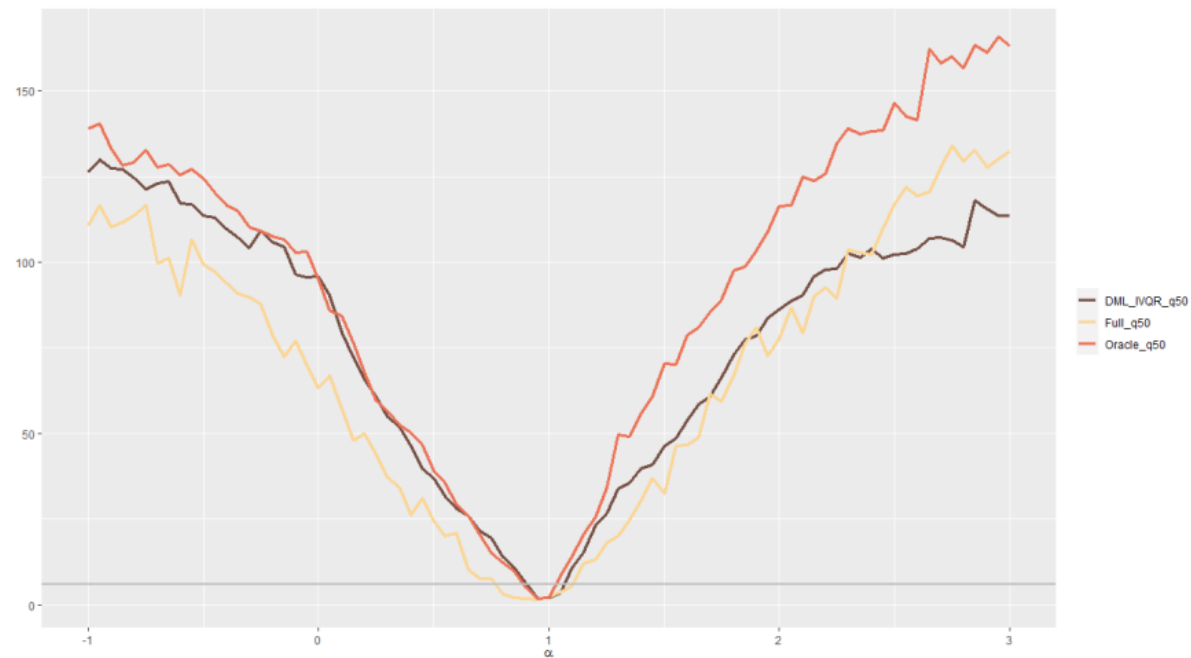


Comparison. Both figures show that DML-IVQR estimates are more concentrated around the true quantile effect than full-GMM estimates, particularly in the tails.

Figure 2: Median, $\tau = 0.50$

Authors

Figure 2. Weak-Instrument Robust Inference at 0.5th quantile:
DML-IVQR (in brown), oracle-GMM, and full-GMM.



Replication

Replication: tau = 0.50

n = 1000; horizontal line = 95% chi-square critical value

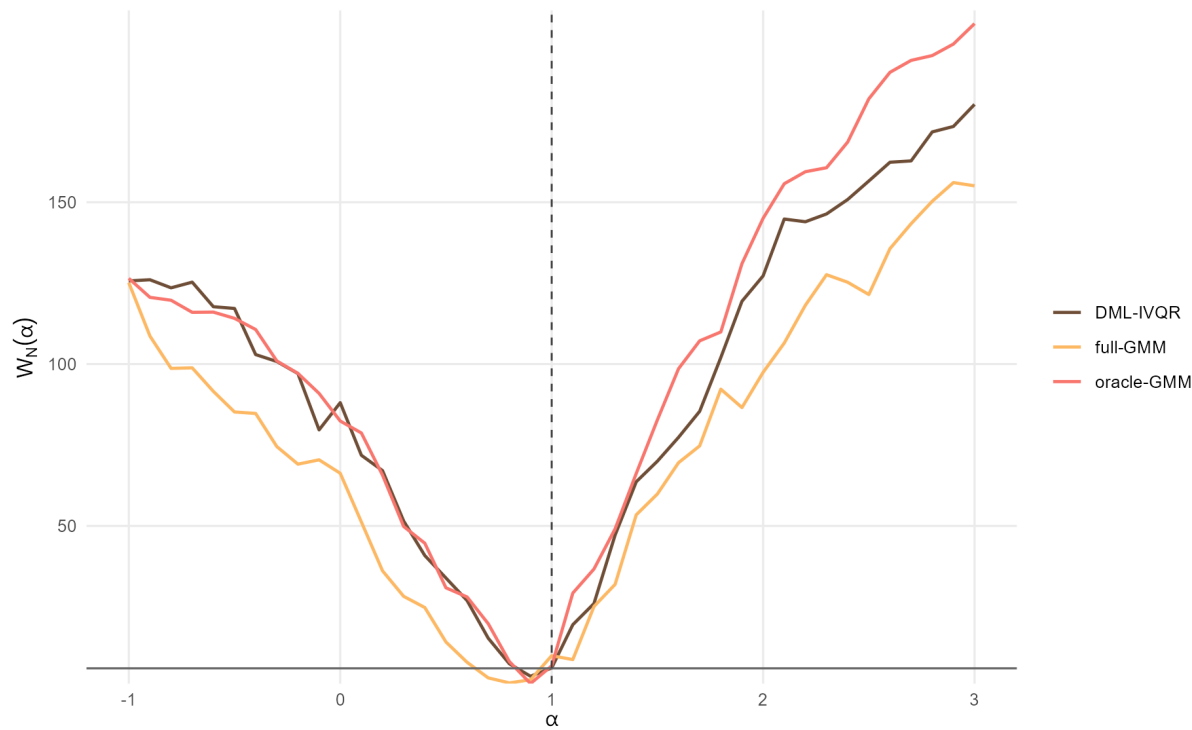
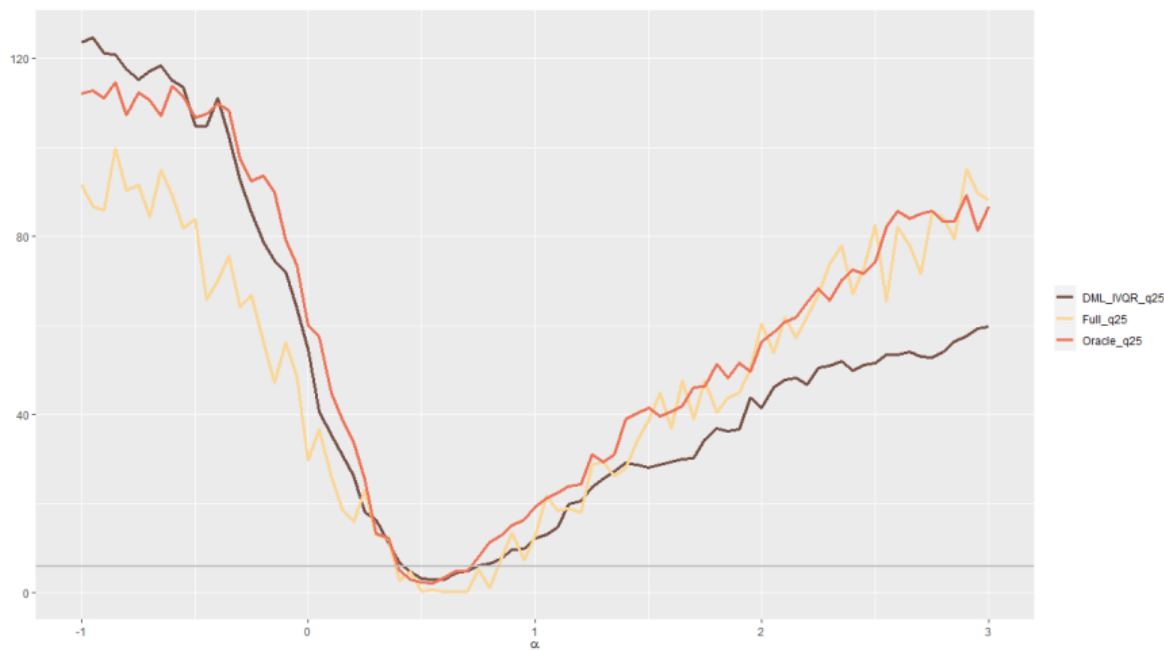


Figure 3: Lower quantile, $\tau = 0.25$

Authors

Figure 3. Weak-Instrument Robust Inference at 0.25th quantile:
DML-IVQR (in brown), oracle-GMM, and full-GMM.



Replication

Replication: tau = 0.25
n = 1000; horizontal line = 95% chi-square critical value

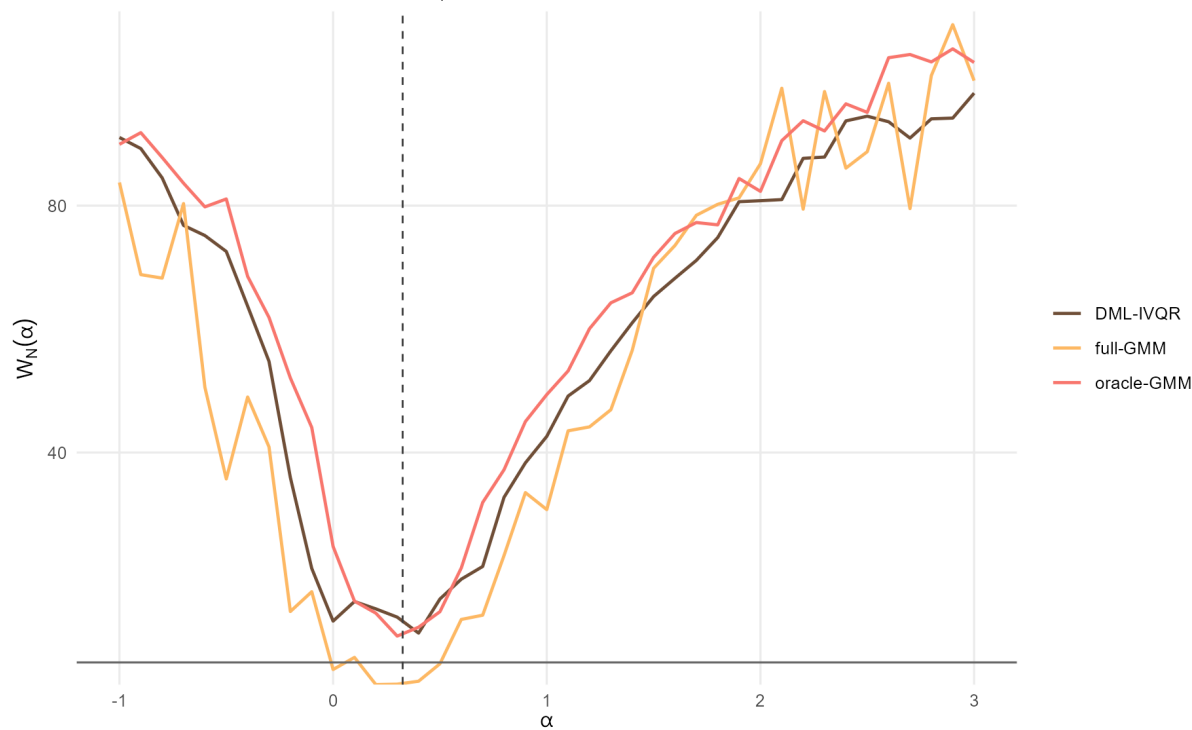
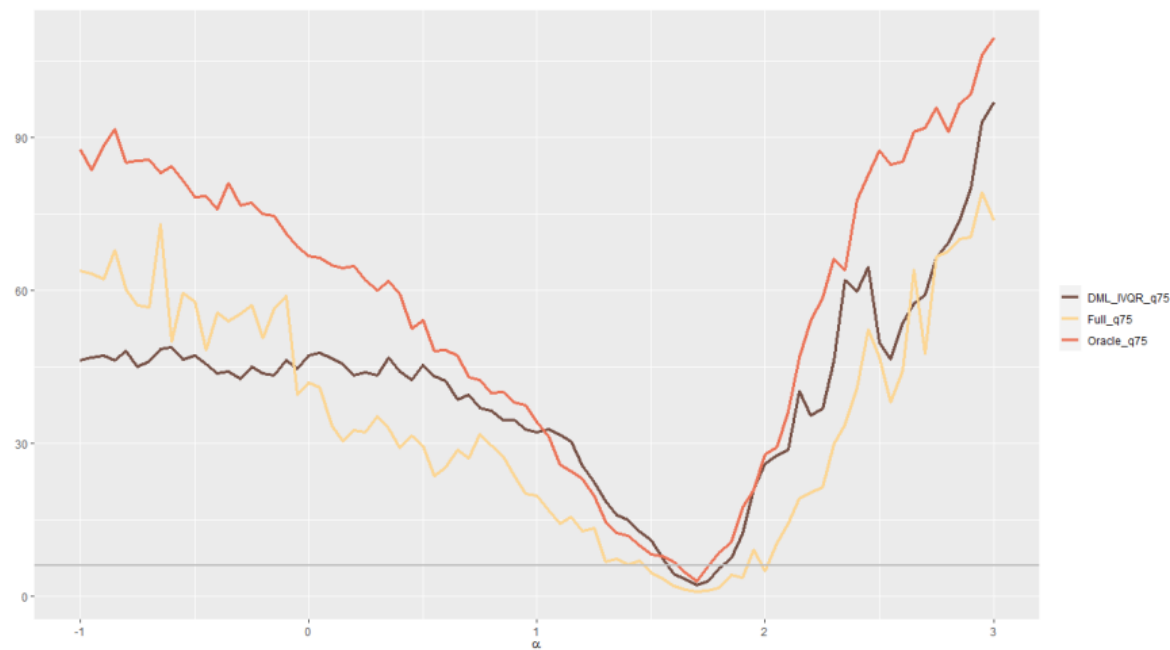


Figure 4: Upper quantile, $\tau = 0.75$

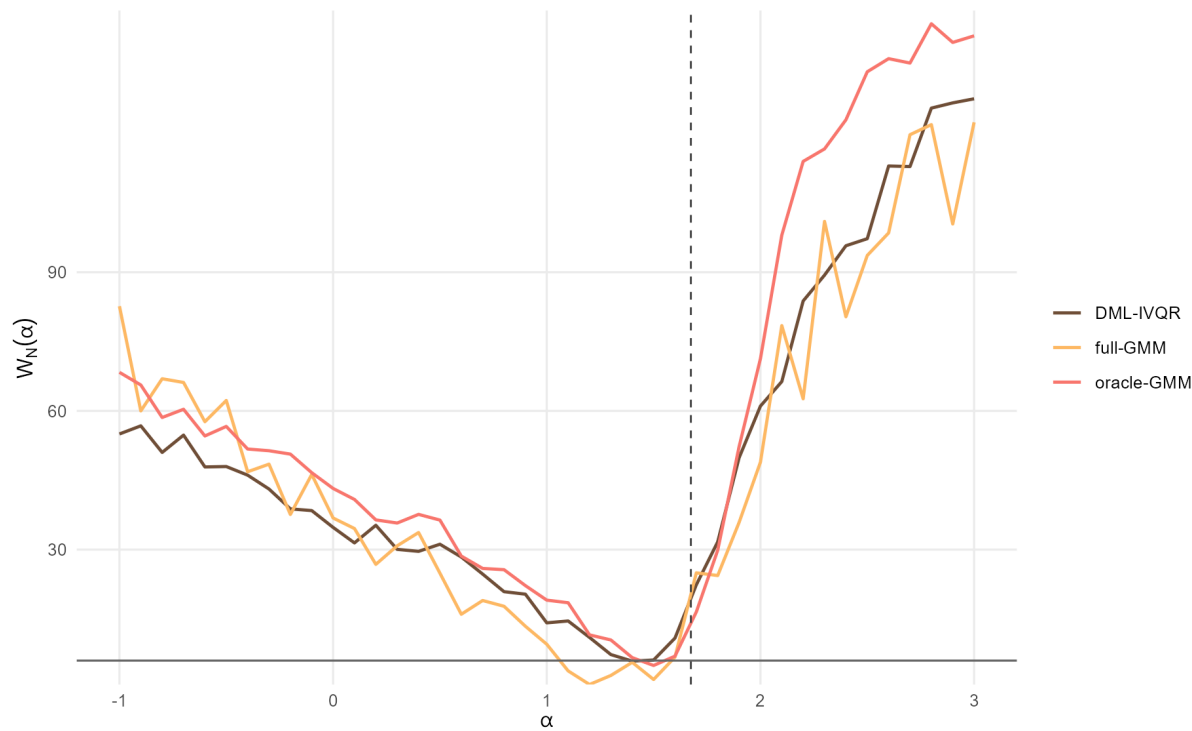
Authors

Figure 4. Weak-Instrument Robust Inference at 0.75th quantile:
DML-IVQR (in brown), oracle-GMM, and full-GMM.



Replication

Replication: tau = 0.75
n = 1000; horizontal line = 95% chi-square critical value



Conclusion

Overall, the Monte Carlo section is successfully replicated. The reproduced results preserve the authors' principal comparative findings.

The authors' main findings were:

1. Residualizing Z on X improves efficiency;
2. Unregularized full-GMM performs poorly with 100 controls;
3. DML-IVQR approaches the oracle benchmark while handling high-dimensional controls;
4. The theoretically motivated and cross-validated penalty choices have broadly similar performance;
5. Weak-identification-robust profile inference remains feasible.

The replication reproduces these qualitative results. Numerical differences remain modest relative to the overall estimator rankings and patterns and are consistent with independent Monte Carlo draws and the newer R/package environment. The strongest evidence comes from Tables 1-3 and Figure 1. Figures 2-4 support the same qualitative inference pattern.